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Inventor(s)

Palmowski; Joseph et al.

SINGLE-ARM ROBOTIC DEVICE WITH COMPACT JOINT DESIGN AND RELATED SYSTEMS AND METHODS

Abstract

Disclosed herein are various robotic surgical devices and systems that include first and second elongate bodies, first and second driveshafts disposed through the second elongate body, and an in-line shoulder joint with a robotic arm coupled thereto. In certain implementations, the in-line shoulder joint has a differential yoke and a dual shaft disposed within the yoke lumen.

Inventors: Palmowski; Joseph (Pasadena, CA), Farritor; Shane (Lincoln, NE)

Applicant: Board of Regents of the University of Nebraska (Lincoln, NE)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application claims priority as a continuation of U.S. application Ser. No. 18/607,863, filed Mar. 18, 2024 and entitled “Single-Arm Robotic Device with Compact Joint Design and Related Systems and Methods,” which is a continuation of U.S. application Ser. No. 17/980,665, filed Nov. 4, 2022, and entitled “Single-Arm Robotic Device with Compact Joint Design and Related Systems and Methods,” which issued as U.S. Pat. No. 11,950,867 on Apr. 9, 2024, which claims priority as a continuation of U.S. application Ser. No. 17/236,489, filed Apr. 21, 2021 and entitled “Single-Manipulator Robotic Device with Compact Joint Design and Related Systems and Methods,” which issued as U.S. Pat. No. 11,504,196 on Nov. 22, 2022, which claims priority as a continuation of U.S. application Ser. No. 16/241,263, filed Jan. 7, 2019 and entitled “Single-Manipulator Robotic Device with Compact Joint Design and Related Systems and Methods,” which issued as U.S. Pat. No. 11,013,564 on May 25, 2021, which claims the benefit under 35 U.S.C. § 119 (e) to U.S. Provisional Application 62/614,127, filed Jan. 5, 2018 and entitled “Single-Manipulator Robotic Device With Compact Joint Design,” the disclosures of which are hereby incorporated herein by reference in their entireties.

FIELD OF THE INVENTION

[0003] The various embodiments herein relate to various medical devices and related components, including robotic and/or in vivo surgical devices and related components, such as arms and end effectors. More specifically, disclosed herein are robotic surgical devices and systems for use in minimally invasive surgical procedures.

BACKGROUND OF THE INVENTION

[0004] Advancements in minimally-invasive surgical (M.I.S.) techniques continue to combat the major drawbacks of open surgery, which include lengthy recovery times, increased chance of postoperative infection, distinct cosmetic remnants, and other such disadvantages. Minimally invasive laparoscopic surgery is one type of M.I.S. technique and consists of the insertion of thin profiled tools and accessories into a gas-inflated abdominal region, eliminating the need for a large incision. However, these procedures also have limitations, including increased duration of surgery, reduction in visibility of the surgical site, and greater dexterity requirements of the surgeon. Accordingly, robotic platform integration has been introduced to the surgical field of medicine and looks to challenge these limitations.

[0005] A more recent minimally-invasive laparoscopic surgical technique is Laparo-Endoscopic Single-Site Surgery (LESS) (also commonly referred to as Single-Incision Laparoscopic Surgery (SILS)), which allows for access through a single incision into the abdomen. In order to operate, the surgeon must cross the tools through the port, causing a loss of triangulation, referred to as the “chopsticks effect.” This process has also included, in some instances, a shift in the type of passageway or access point into the peritoneal cavity from the traditional trocar to a single-port entry device.

[0006] While single-port entry systems for use in LESS/SILS procedures are currently

insubstantial, one device that has gained popularity is the GelPort® (hereafter referred to as “GelPort”) system, which has been adapted to work as a passageway for trocars and robotic platforms.

[0007] By comparison, a traditional trocar is a relatively simple device that includes two known components: an outer sleeve (or “cannula”) and an obturator. The tube-like cannula acts as the actual passageway into the abdomen, while the obturator is the component with the generally sharp distal end that is positioned through the cannula and creates the passageway into the peritoneal cavity via the skin of the patient.

[0008] The industry standard in externally-actuated surgical robotic platforms is the Da Vinci® Surgical System (hereafter referred to as “Da Vinci”), which consists of a master-slave configuration in which the surgeon(s) operates from a remote console. Da Vinci is also capable of performing single-site surgery. Disadvantages of Da Vinci include cost (the system costs millions of dollars, making it unaffordable for smaller less funded hospitals), having a large base that takes up a good portion of the area surrounding the operating bed (as well as various additional consoles scattered throughout the operating room), and being a highly complex system that requires extensive training and experience before a surgeon can operate on a human.

[0009] There is a need in the art for an improved robotic surgical system.

BRIEF SUMMARY OF THE INVENTION

[0010] Discussed herein are various single-armed robotic device embodiments with in-line shoulder joints.

[0011] In Example 1, a robotic device comprises a first elongate device body comprising first and second motors, a second elongate device body coupled to a distal end of the first elongate device body, a first driveshaft disposed through the second elongate device body and operably coupled to the first motor, a second driveshaft disposed through the second elongate device body and operably coupled to the second motor, the second driveshaft operably coupled at a distal end to a first bevel gear, a shoulder joint, and an arm operably coupled to the output body. The shoulder joint comprises a differential yoke rotationally coupled to the first driveshaft, a dual shaft rotatably disposed within the yoke lumen, and an output body rotatably disposed over the extension shaft. The differential yoke comprises a yoke body and a yoke lumen defined in the differential yoke, wherein the lumen has a longitudinal axis that is transverse to the longitudinal axis of the body. In addition, the dual shaft comprises a rotational shaft rotatably disposed within the yoke lumen, the rotational shaft rotationally coupled to the first bevel gear, and an extension shaft extending from the rotational shaft such that a longitudinal axis of the extension shaft is transverse to a longitudinal axis of the rotational shaft.

[0012] Example 2 relates to the robotic device according to Example 1, wherein the first driveshaft is rotatably disposed within and being radially concentric with the second driveshaft.

[0013] Example 3 relates to the robotic device according to Example 1, further comprising a third driveshaft operably coupled to a third motor, the third driveshaft operably coupled at a distal end to a second bevel gear, wherein the second bevel gear is operably coupled to the output body.

[0014] Example 4 relates to the robotic device according to Example 3, wherein the first driveshaft is rotatably disposed within and being radially concentric with the third driveshaft, and wherein the third driveshaft is rotatably disposed within and being radially concentric with the second driveshaft.

[0015] Example 5 relates to the robotic device according to Example 1, wherein the device has no second arm.

[0016] Example 6 relates to the robotic device according to Example 1, wherein the shoulder joint is a unitary shoulder joint.

[0017] Example 7 relates to the robotic device according to Example 1, wherein the second elongate device body has a cross-sectional diameter that is less than a cross-sectional diameter of the first elongate device body.

[0018] Example 8 relates to the robotic device according to Example 1, wherein the second elongate device body is sized and structured to be positionable through a standard trocar port.

[0019] Example 9 relates to the robotic device according to Example 1, wherein the shoulder joint is an in-line joint that is collinear with the second elongate device body.

[0020] In Example 10, a robotic device comprises a first elongate device body comprising first, second, and third motors disposed within the first elongate device body, a second elongate device body coupled to a distal end of the first elongate device body, a first driveshaft disposed through the second elongate device body and operably coupled to the first motor, a second driveshaft disposed through the second elongate device body and operably coupled to the second motor, the second driveshaft operably coupled at a distal end to a first bevel gear, a third driveshaft disposed through the second elongate device body and operably coupled to the third motor, the third driveshaft operably coupled at a distal end to a second bevel gear, a shoulder joint, and an arm operably coupled to the output body. The shoulder joint comprises a differential yoke rotationally coupled to the first driveshaft, a T shaft rotatably disposed within the yoke lumen, and an output body rotatably disposed over the extension shaft, the output body rotationally coupled to the first bevel gear. The differential yoke comprises a yoke body and a yoke lumen defined in the differential yoke, wherein the yoke lumen has a longitudinal axis that is transverse to the longitudinal axis of the body. The T shaft comprises a rotational shaft rotatably disposed within the yoke lumen, the rotational shaft rotationally coupled to the second bevel gear, and an extension shaft extending from the rotational shaft.

[0021] Example 11 relates to the robotic device according to Example 10, wherein the first driveshaft is rotatably disposed within and radially concentric with the second driveshaft, and wherein the second driveshaft is rotatably disposed within and radially concentric with the third driveshaft.

[0022] Example 12 relates to the robotic device according to Example 10, wherein the yoke lumen is defined by opposing first and second yoke lumen walls, the first yoke lumen wall comprising a first slot and the second yoke lumen wall comprising a second slot.

[0023] Example 13 relates to the robotic device according to Example 12, wherein the extension shaft is disposable within the first and second slots.

[0024] Example 14 relates to the robotic device according to Example 10, wherein a longitudinal axis of the extension shaft is transverse to a longitudinal axis of the rotational shaft.

[0025] Example 15 relates to the robotic device according to Example 10, wherein the first elongate device body has a cross-sectional diameter that is greater than a cross-sectional diameter of the second elongate device body.

[0026] Example 16 relates to the robotic device according to Example 10, wherein the arm comprises an upper arm body; a forearm body; an elbow joint coupling the forearm body to the upper arm body; an end effector; and a wrist joint coupling the end effector to the forearm body, wherein the upper arm body is operably coupled to the output body.

[0027] In Example 17, a robotic device comprises a first elongate device body comprising first, second, and third motors disposed within the first elongate device body, a second elongate device body coupled to a distal end of the first elongate device body, a first driveshaft disposed through the second elongate device body and operably coupled to the first motor, a second driveshaft disposed through the second elongate device body and operably coupled to the second motor, a third driveshaft disposed through the second elongate device body and operably coupled to the third motor, a shoulder joint, and an arm operably coupled to the output body. The second driveshaft is operably coupled at a distal end to a first bevel gear, wherein the first driveshaft is rotatably disposed within and radially concentric with the second driveshaft. The third driveshaft is operably coupled at a distal end to a second bevel gear, wherein the second driveshaft is rotatably disposed within and radially concentric with the third driveshaft. The shoulder joint comprises a differential yoke rotationally coupled to the first driveshaft, a T shaft rotatably disposed within the

yoke lumen, and an output body rotatably disposed over the extension shaft, the output body rotationally coupled to the first bevel gear. The differential yoke comprises a yoke body and a yoke lumen defined in the differential yoke by opposing first and second curved shells, wherein the first curved shell comprises a first slot and the second curved shell comprises a second slot, and wherein the yoke lumen has a longitudinal axis that is transverse to the longitudinal axis of the body. The T shaft comprises a rotational shaft rotatably disposed within the yoke lumen, the rotational shaft rotationally coupled to the second bevel gear, and an extension shaft extending from the rotational shaft.

[0028] Example 18 relates to the robotic device according to Example 17, wherein the extension shaft is disposable within the first and second slots.

[0029] Example 19 relates to the robotic device according to Example 17, wherein the first elongate device body has a cross-sectional diameter that is greater than a cross-sectional diameter of the second elongate device body.

[0030] Example 20 relates to the robotic device according to Example 17, wherein the shoulder joint has a cross-sectional diameter that is less than or substantially similar to a cross-sectional diameter of the second elongate body.

[0031] While multiple embodiments are disclosed, still other embodiments of the present invention will become apparent to those skilled in the art from the following detailed description, which shows and describes illustrative embodiments of the invention. As will be realized, the invention is capable of modifications in various obvious aspects, all without departing from the spirit and scope of the present invention. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] FIG. 1A is a perspective view of a single-armed robotic device disposed through a trocar port, according to one embodiment.

[0033] FIG. 1B is a side view of the robotic device of FIG. 1A, according to one embodiment.

[0034] FIG. 1C is a top view of the robotic device of FIG. 1A, according to one embodiment.

[0035] FIG. 2A is a side view of the first and second elongate bodies of a robotic device, according to one embodiment.

[0036] FIG. 2B is a cross-sectional cutaway side view of a portion of the first and second elongate bodies of the robotic device of FIG. 2A, according to one embodiment.

[0037] FIG. 2C is a cross-sectional, partial cutaway perspective view of an expanded portion of the first and second elongate bodies of the robotic device of FIG. 2A, according to one embodiment.

[0038] FIG. 2D is a cross-sectional, partial cutaway perspective view of an expanded portion of the second elongate body and shoulder joint of the robotic device of FIG. 2A, according to one embodiment.

[0039] FIG. 2E is a cross-sectional cutaway bottom view of the shoulder joint of the robotic device of FIG. 2A, according to one embodiment.

[0040] FIG. 2F is an exploded perspective view of the shoulder joint of the robotic device of FIG. 2A, according to one embodiment.

[0041] FIG. 3 is a cross-sectional cutaway top view of a shoulder joint of a robotic device, according to one embodiment.

[0042] FIG. 4 is a perspective view of a robotic arm and a portion of a second elongate body of a single-armed robotic device, according to one embodiment.

[0043] FIG. 5A is a side view of an upper arm of a robotic arm of a robotic device, according to one embodiment.

[0044] FIG. 5B is a cross-sectional cutaway side view of the elbow joint of the robotic arm of FIG. 5A, according to one embodiment.

[0045] FIG. 6A is a perspective view of a forearm of a robotic arm, according to one embodiment.

[0046] FIG. 6B is a cross-sectional, partial cutaway perspective view of an expanded portion of the forearm of FIG. 6A, according to one embodiment.

[0047] FIG. 6C is another cross-sectional, partial cutaway perspective view of an expanded portion of the forearm of FIG. 6A, according to one embodiment.

[0048] FIG. 7A is a side view of the forearm of FIG. 6A, according to one embodiment.

[0049] FIG. 7B is a cross-sectional cutaway side view of a portion of the forearm and the wrist joint of the forearm of FIG. 6A, according to one embodiment.

[0050] FIG. 8A is a perspective view of a laparoscopic trocar, according to one embodiment.

[0051] FIG. 8B is a side view of the laparoscopic trocar of FIG. 8A, according to one embodiment.

[0052] FIG. 9A is a perspective exterior view of a one-armed robotic device positioned within a target cavity of a patient, according to one embodiment.

[0053] FIG. 9B is a schematic view of the calculated workspace of the one-armed robotic device of FIG. 9A, according to one embodiment.

DETAILED DESCRIPTION

[0054] The various robotic surgical device and system embodiments disclosed or contemplated herein include a robotic device having an elongate body having a minimal cross-sectional profile and a single robotic arm attached thereto via an in-line shoulder joint.

[0055] It is understood that the various implementations of robotic devices and related methods and systems disclosed herein can be incorporated into or used with any other known medical devices, systems, and methods. For example, the various implementations disclosed herein may be incorporated into or used with any of the medical devices and systems disclosed in U.S. Pat. No. 7,492,116 (filed on Oct. 31, 2007 and entitled “Robot for Surgical Applications”), 7,772,796 (filed on Apr. 3, 2007 and entitled “Robot for Surgical Applications”), U.S. Pat. No. 8,179,073 (issued May 15, 2011, and entitled “Robotic Devices with Agent Delivery Components and Related Methods”), 8,343, 171 (issued Jan. 1, 2013 and entitled “Methods and Systems of Actuation in Robotic Devices”), U.S. Pat. No. 8,679,096 (issued Mar. 25, 2014 and entitled “Multifunctional Operational Component for Robotic Devices”), U.S. Pat. No. 8,828,024 (issued on Sep. 9, 2014 and entitled “Methods and Systems of Actuation in Robotic Devices”), U.S. Pat. No. 8,834,488 (issued Sep. 16, 2014 and entitled “Magnetically Coupleable Surgical Robotic Devices and Related Methods”), U.S. Pat. No. 8,894,633 (issued Nov. 25, 2014 and entitled “Modular and Cooperative Medical Devices and Related Systems and Methods”), U.S. Pat. No. 8,968,267 (issued Mar. 3, 2015 and entitled “Methods and Systems for Handling or Delivering Materials for Natural Orifice Surgery”), U.S. Pat. No. 8,968,332 (issued Mar. 3, 2015 and entitled “Magnetically Coupleable Robotic Devices and Related Methods”), U.S. Pat. No. 8,974,440 (issued Mar. 10, 2015 and entitled “Modular and Cooperative Medical Devices and Related Systems and Methods”), U.S. Pat. No. 9,010,214 (Apr. 21, 2015 and entitled “Local Control Robotic Surgical Devices and Related Methods”), U.S. Pat. No. 9,060,781 (issued Jun. 23, 2015 and entitled “Methods, Systems, and Devices Relating to Surgical End Effectors”), U.S. Pat. No. 9,089,353 (issued Jul. 28, 2015 and entitled “Robotic Surgical Devices, Systems, and Related Methods”), U.S. Pat. No. 9,179,981 (issued on Nov. 10, 2015 and entitled “Multifunctional Operational Component for Robotic Devices”), U.S. Pat. No. 9,498,292 (issued Nov. 22, 2016 and entitled “Single Site Robotic Devices and Related Systems and Methods”), U.S. Pat. No. 9,579,088 (issued Feb. 28, 2017 and entitled “Methods, Systems, and Devices for Surgical Visualization and Device Manipulation”), U.S. Pat. No. 9,743,987 (Aug. 29, 2017 and entitled “Methods, Systems, and Devices Relating to Robotic Surgical Devices, End Effectors, and Controllers”), U.S. Pat. No. 9,770,305 (issued Sep. 26, 2017 and entitled “Robotic Surgical Devices, Systems, and Related Methods”), and U.S. Pat. No. 9,888,966 (issued Feb. 13, 2018 and entitled “Methods, Systems, and Devices Relating to Force

Control Surgical Systems), all of which are hereby incorporated herein by reference in their entireties.

[0056] Further, the various implementations disclosed herein may be incorporated into or used with any of the medical devices and systems disclosed in copending U.S. Published Applications 2014/0046340 (filed Mar. 15, 2013 and entitled “Robotic Surgical Devices, Systems, and Related Methods”), 2014/0058205 (filed Jan. 10, 2013 and entitled “Methods, Systems, and Devices for Surgical Access and Insertion”), 2014/0303434 (filed Mar. 14, 2014 and entitled “Robotic Surgical Devices, Systems, and Related Methods”), 2015/0051446 (filed Jul. 17, 2014 and entitled “Robotic Surgical Devices, Systems, and Related Methods”), 2016/0074120 (filed Sep. 14, 2015, and entitled “Quick-Release End Effectors and Related Systems and Methods”), 2016/0135898 (filed Nov. 11, 2015 entitled “Robotic Device with Compact Joint Design and Related Systems and Methods”), 2016/0157709 (filed Feb. 8, 2016 and entitled “Medical Inflation, Attachment, and Delivery Devices and Related Methods”), 2017/0035526 (filed Aug. 3, 2016 and entitled “Robotic Surgical Devices, Systems, and Related Methods”), 2017/0354470 (filed May 18, 2017 and entitled “Robotic Surgical Devices, Systems, and Related Methods”), 2018/0055584 (filed Aug. 30, 2017 and entitled “Robotic Device with Compact Joint Design and an Additional Degree of Freedom and Related Systems and Methods”), 2018/0056527 (filed Aug. 25, 2017 and entitled “Quick-Release End Effector Tool Interface”), 2018/0140377 (filed Nov. 22, 2017 and entitled “Gross Positioning Device and Related Systems and Methods”), 2018/0147019 (filed Nov. 29, 2017 and entitled “User Controller with User Presence Detection and Related Systems and Methods”), and 2018/0161122 (filed Dec. 14, 2017 and entitled “Releasable Attachment Device for Coupling to Medical Devices and Related Systems and Methods”), all of which are hereby incorporated herein by reference in their entireties. In addition, the various implementations disclosed herein may be incorporated into or used with any of the medical devices and systems disclosed in copending U.S. application Ser. No. 16/144,807 (filed Sep. 27, 2018), which is hereby incorporated herein by reference in its entirety.

[0057] Certain device and system implementations disclosed in the patents and/or applications listed above can be positioned within a body cavity of a patient in combination with a support component similar to those disclosed herein and/or in certain of the patents and/or applications incorporated by reference above. An “in vivo device” as used herein means any device that can be positioned, operated, or controlled at least in part by a user while being positioned within a body cavity of a patient, including any device that is disposed through an opening or orifice of the body cavity and is coupled to a support component such as a rod or other such component, also including any device positioned substantially against or adjacent to a wall of a body cavity of a patient, further including any such device that is internally actuated (having no external source of motive force), and additionally including any device that may be used laparoscopically or endoscopically during a surgical procedure. As used herein, the terms “robot,” and “robotic device” shall refer to any device that can perform a task either automatically or in response to a command.

[0058] Certain implementations provide for insertion of the present invention into the cavity while maintaining sufficient insufflation of the cavity. Further implementations minimize the physical contact of the surgeon or surgical users with the present invention during the insertion process. Other implementations enhance the safety of the insertion process for the patient and the present invention. For example, some implementations provide visualization of the present invention as it is being inserted into the patient's cavity to ensure that no damaging contact occurs between the system/device and the patient. In addition, certain implementations allow for minimization of the incision size/length. Further implementations reduce the complexity of the access/insertion procedure and/or the steps required for the procedure. Other implementations relate to devices that have minimal profiles, minimal size, or are generally minimal in function and appearance to enhance ease of handling and use.

[0059] As in manual laparoscopic procedures, a known insufflation system can be used to pump

sterile carbon dioxide (or other gas) into the patient's abdominal cavity. This lifts the abdominal wall from the organs and creates space for the robot. In certain implementations, the system has no direct interface with the insufflation system. Alternatively, the system can have a direct interface to the insufflation system.

[0060] According to various embodiments, the insertion port is a traditional trocar or a modified trocar as described elsewhere herein. In such implementations, the device embodiments herein—or a portion thereof—are disposed through the cannula of the trocar in a fashion described in more detail below. Alternatively, in certain implementations, the insertion port is single-port entry device that is a known, commercially-available flexible membrane placed transabdominally to seal and protect the abdominal incision. One example of such a single-port entry device is the GelPort® discussed above. This off-the-shelf component is the same device used in the same way for Hand-Assisted Laparoscopic Surgery (HALS). The only difference is that the working arms of the robot are inserted into the abdominal cavity through the insertion port rather than the surgeon's hand. The robot body seals against the insertion port, thereby maintaining insufflation pressure. The port is single-use and disposable. In a further alternative, any known port can be used.

[0061] Certain implementations disclosed herein relate to “combination” or “modular” medical devices that can be assembled in a variety of configurations. For purposes of this application, both “combination device” and “modular device” shall mean any medical device having modular or interchangeable components that can be arranged in a variety of different configurations, and the related systems. The modular components and combination devices disclosed herein also include segmented triangular or quadrangular-shaped combination devices. These devices, which are made up of modular components (also referred to herein as “segments”) that are connected to create the triangular or quadrangular configuration, can provide leverage and/or stability during use while also providing for substantial payload space within the device that can be used for larger components or more operational components. As with the various combination devices disclosed and discussed above, according to one implementation these triangular or quadrangular devices can be positioned inside the body cavity of a patient in the same fashion as those devices discussed and disclosed above.

[0062] The various system implementations described herein are used to perform robotic surgery. Further, the various implementations disclosed herein can be used in a minimally invasive approach to a variety of procedures that are typically performed “open” by known technologies, with the potential to improve clinical outcomes and health care costs, including, for example, general surgery applications in the abdominal cavity, such as, for example, colon resection and other known procedures. Further, the various implementations disclosed herein can be used in place of the known mainframe-like laparoscopic surgical robots that reach into the body from outside the patient. That is, the less-invasive robotic systems, methods, and devices according to the implementations disclosed herein feature small, self-contained surgical devices that are inserted in their entireties through a single incision in the patient's abdomen. Designed to utilize existing tools and techniques familiar to surgeons, the devices disclosed herein will not require a dedicated operating room or specialized infrastructure, and, because of their much smaller size, are expected to be significantly less expensive than existing robotic alternatives for laparoscopic surgery. Due to these technological advances, the various implementations herein could enable a minimally invasive approach to procedures performed in open surgery today. In certain implementations, the various systems described herein are based on and/or utilize techniques used in manual laparoscopic surgery including insufflation of the abdominal cavity and the use of ports to insert tools into the abdominal cavity.

[0063] As will be described in additional detail below, components of the various system implementations disclosed or contemplated herein can include a control console and a robot having a minimal cross-sectional profile and a single robotic arm as described herein. The robot implementations are constructed and arranged to be inserted into the insufflated abdominal cavity

as described in further detail herein.

[0064] While the various embodiments herein are described as having motors, it is understood that any type of known actuators can be used in place of the motors. Further, while the transmission components that transmit the force from the motors within the device are described as shafts that are coupled via gears, it is understood that any type of transmission components or mechanisms can be incorporated herein in place of such shafts and gears, including hydraulic or pneumatic components or other mechanical components such as cables and pulleys or similar components.

[0065] The various implementations are disclosed in additional detail in the attached figures and the related discussion thereof, as set forth herein.

[0066] FIGS. **1A-1C** depict one embodiment of the single manipulator device (also referred to herein as the “unitary manipulator,” “single arm,” “unitary arm,” “in-line joint,” “in-line shoulder joint,” and “robotic surgical” device) **10** according to one embodiment. In this implementation, the device **10** has a first body (also referred to herein as an “external body” or “first support beam”) **12**, a second body (also referred to herein as an “internal body” or “second support beam”) **14**, a first arm component (also referred to herein as an “upper arm”) **16**, a second arm component (also referred to herein as a “forearm”) **18**, and an operational component (also referred to herein as an “end effector”) **20**. In this embodiment as depicted and various other implementations herein, the internal body **14** has a smaller cross-sectional diameter than the external body **12** such that the internal body **14** can be inserted through a smaller port than would be possible for the external body **12**, including, for example, a trocar port such as the port **28** depicted in FIG. **1A**. Note in FIG. **1A** that the internal body **14** is disposed through and positioned within the trocar port **28** such that the distal end of the internal body **14** extends out of the distal end of the trocar cannula **30**. As such, the distal end of the internal body **14** and the shoulder joint **22** are disposed within the internal target cavity of the patient. The second body **14** is coupled to the first body **12**. In certain embodiments, the first and second bodies **12**, **14** are removably coupled. Alternatively, the first and second bodies **12**, **14** are fixedly coupled or are integral with each other as a single, unitary component. In a further alternative, the first and second bodies **12**, **14** are moveably coupled to each other in a jointed or other non-rigid fashion.

[0067] The first arm component **16** is rotatably coupled to the second body **14** via a first joint (also referred to herein as a “shoulder joint”) **22**, while the second arm component **18** is rotatably coupled to the first arm component **16** via a third joint (also referred to herein as an “elbow joint”) **24**, and the operational component **20** is rotatably coupled to the second arm component **18** via a fourth joint (also referred to herein as a “wrist joint”) **26**. Each of these components will be discussed in further detail below, according to various embodiments of the robotic surgical device.

[0068] It is understood that the compact in-line (or “coaxial”) shoulder joint as described in additional detail with respect to the various embodiments disclosed or contemplated herein presents a less-invasive approach to the insertable portion of the robot in comparison to devices having shoulder joints that are not in-line, including those devices having two shoulder joints. That is, the in-line shoulder joint implementations herein have a smaller cross-sectional diameter in comparison to the known non-in-line or double shoulder joints.

[0069] One embodiment of a single-armed device **40**, and more specifically the external body **42**, the internal body **44**, and shoulder joint **46** thereof, is depicted in further detail in FIGS. **2A-2F**. Like every other implementation disclosed or contemplated herein, this specific device **40** embodiment has a shoulder joint **46** with 3 degrees of freedom (“DOF”) and a minimal cross-sectional diameter for ease of insertion through trocars and other insertion devices. The minimal diameter is accomplished via the specific design of the various components within the internal body **44** and the joint **46**, as described in further detail below.

[0070] FIG. **2A** depicts a side view of the device **40**, including the external body **42**, the internal body **44**, and the shoulder joint **46**. More specifically, the figure shows the external casing or housing **48** of the external body **42** and the external casing or housing **50** of the internal body **44**.

[0071] FIG. 2B depicts a cross-sectional front view of the external body **42**, the internal body **44**, and the shoulder joint **46** in which certain internal components of those components are visible, according to one exemplary embodiment. In this implementation and other embodiments disclosed or contemplated herein, the external body **42** contains the actuators (which, in this specific embodiment are motors) **70A**, **70B**, **70C** that actuate the shoulder joint **46** via the transmission shafts (also referred to herein as “nested driveshafts”) **52** disposed within the internal body **44**. This configuration (with the motors **70A**, **70B**, **70C** disposed within the external body **42** and the nested driveshafts **52** disposed in a nested configuration within the internal body **44** makes it possible for the internal body **44** to have a smaller cross-sectional diameter than the external body **42**, as discussed above with respect to device **10**. The set of nested driveshafts **52** are rotatably disposed within the internal body **44**. As set forth herein, the word “nested” is intended to describe components that are concentric such that at least one of the components is positioned inside another of those components and each of the components have a common axis of rotation.

[0072] With respect to FIG. 2B, the set of nested driveshafts **52** is made up of a first or outer driveshaft **52A**, a second or middle driveshaft **52B**, and a third or inner driveshaft **52C**. The set of nested driveshafts **52** extend from the external body **42** into and through the internal body **44** as shown. The inner driveshaft **52C** is rotatably disposed within the middle driveshaft **52B** as shown, and has a driven gear **54C** fixedly or integrally attached at its proximal end. At its distal end, the inner driveshaft **52C** is coupled to or integral with a differential yoke (also referred to herein as a “shoulder housing” or “conversion body”) **56C**. The middle driveshaft **52B** is rotatably disposed within the outer driveshaft **52A** as shown, and has a driven gear **54B** fixedly or integrally attached at its proximal end. At its distal end, the middle driveshaft **52B** is coupled to a second or inner drive bevel gear **56B**. The outer driveshaft **52A** is rotatably disposed within the internal body **44** and has a driven gear **54A** fixedly or integrally attached at its proximal end. At its distal end, the outer driveshaft **52A** is coupled to a first or outer drive bevel gear **56A**.

[0073] Various internal components at the proximal end of the internal body **44** and distal end of the external body **42** are best depicted in FIGS. 2B and 2C, including the proximal end of the driveshafts **52A**, **52B**, **52C** and related gears and motors that drive those driveshafts **52A**, **52B**, **52C**, according to one embodiment. As best shown in FIG. 2B, the proximal end of the inner driveshaft **52C** is rotatably supported in the external body **42** via a first shaft bearing **60** and a second shaft bearing **62**. Further, the proximal end of the middle driveshaft **52B**, including the driven gear **54B**, is rotatably supported in the external body **42** via the second shaft bearing **62** and a third shaft bearing **64**. In addition, the proximal end of the outer driveshaft **52A**, including driven gear **54A**, is rotatably supported in the external body **42** via the third shaft bearing **64**.

[0074] As best shown in FIGS. 2B and 2C, the set of nested driveshafts **52** has three motors operably coupled thereto (although only two of the three motors are visible), wherein the three motors are disposed within the external body **42** as discussed above. More specifically, motor **70A** has a motor drive gear **72A** that is coupled to the driven gear **54A** (which is coupled to the outer driveshaft **52A**). Further, motor **70C** has a motor drive gear **72C** that is coupled to the driven gear **54C** (which is coupled to the inner driveshaft **52C**). In addition, a third motor (logically identified as **70B** but not visible in the figures due to the perspective thereof) has a third motor drive gear (logically identified as **72B** but also not visible) that is coupled to the driven gear **54B** (which is coupled to the middle driveshaft **52B**). The positioning and functionality of the third motor and third motor drive gear are understood by one of ordinary skill in the art based on the description herein.

[0075] In one embodiment, the motors **70A**, **70B**, **70C** are 12 volt Faulhaber **1226** series brushless DC motors coupled to Faulhaber **256:1** planetary gearboxes with an efficiency rating of 0.60. Alternatively, any known motors can be used in the external body **42**.

[0076] Thus, in operation, the motor **70A** can be actuated to drive rotation of the outer driveshaft **52A** by driving rotation of motor drive gear **72A**, which drives rotation of the driven gear **54A**.

Similarly, the motor **70B** (not visible in the figures) can be actuated to drive rotation of the middle driveshaft **52B** by driving rotation of motor drive gear **72B** (also not visible), which drives rotation of the driven gear **54B**. In a similar fashion, the motor **70C** can be actuated to drive rotation of the inner driveshaft **52C** by driving rotation of motor drive gear **72C**, which drives rotation of the driven gear **54C**.

[0077] FIGS. **2B**, **2D**, **2E**, and **2F** depict the shoulder joint **46** and its various components, according to one implementation. More specifically, FIG. **2D** depicts a cross-sectional perspective view of the internal components of the joint **46**, while FIG. **2E** depicts a cross-sectional top view of the joint **46**, and FIG. **2F** depicts an exploded view of the internal components of the joint **46**. As discussed above, and as shown in FIGS. **2B**, **2D**, and **2F**, the outer driveshaft **52A** is coupled (or rotationally constrained) to the outer drive bevel gear **56A**, while the middle driveshaft **52B** is coupled to the inner drive bevel gear **56B**, and the inner driveshaft **52C** is coupled to the differential yoke **56C**. As best shown in FIG. **2B**, at the distal end, the outer driveshaft **52A** and outer drive bevel gear **56A** are supported by the first shoulder bearing **80**. In addition, the middle driveshaft **52B** and the inner drive bevel gear **56B** are supported by the first shoulder bearing **80** and the second shoulder bearing **82**.

[0078] As best shown in FIGS. **2D** and **2F**, the differential yoke **56C** has a cylindrical body **90** and a yoke opening or partial lumen **93** defined at the distal end of the body **90**. In one implementation as shown, the lumen **93** is defined by two opposing curved shells (or “curved walls”) **92A**, **92B** that extend from the body **90** and define the yoke lumen **93** between the two shells **92A**, **92B**. In one embodiment, each of the two shells **92A**, **92B** is a set of two prongs, such that the first shell **92A** is a first set of two curved prongs **92A** and the second shell **92B** is a second set of two curved prongs **92B**. In addition, shaft slots **94A**, **94B** are defined in the two shells **92A**, **92B**. Thus, in those embodiments in which the shells **92A**, **92B** are prong pairs **92A**, **92B**, the shaft slots **94A**, **94B** are defined between each of two prongs of each set of prongs **92A**, **92B**. The cylindrical body **90** is disposed in the joint **46** such that its longitudinal axis is parallel to and concentric with the longitudinal axis of the internal body **44**. In contrast, the yoke lumen **93** defined by the two shells **92A**, **92B** has a longitudinal axis that is transverse to the longitudinal axis of the cylindrical body **90**.

[0079] Further, the joint **46** has a dual shaft (also referred to herein as a “t-bar” or “T shaft”) **96** rotatably disposed within the yoke opening **93**. The dual shaft **96** has a rotational shaft (also referred to as the “main shaft” or “pitch shaft”) **96A** and an extension shaft (also referred to as the “roll shaft”) **96B**. The extension shaft **96B** has a longitudinal axis that is transverse to the longitudinal axis of the rotational shaft **96A**. The pitch shaft **96A** is rotatably supported within the yoke opening **93** by the third **84** and fourth **86** shoulder bearings (as best shown in FIG. **2B**)—along with the fifth **88** and sixth **89** shoulder bearings (also best shown in FIG. **2B**)—such that the pitch shaft **96A** rotates around a longitudinal axis that is transverse to the longitudinal axis of the internal body **44** (and thus also the longitudinal axes of the set of nested driveshafts **52**). Further, the pitch shaft **96A** is coupled or rotationally constrained to the first shoulder bevel gear **100** such that rotation of the first shoulder gear **100** causes rotation of the rotational shaft **96A**. The first shoulder gear **100** is rotatably coupled to the outer drive bevel gear **56A** such that rotation of the outer drive bevel gear **56A** causes rotation of the first shoulder gear **100**. The bevel gear **100**, the pitch shaft **96A**, and the bearings **84**, **86**, **88**, **89** are coupled together and “preloaded” by the screw **102** that is coupled to the pitch shaft **96A**. Alternatively, any known attachment component can be used to couple together and preload these components. As such, rotation of the motor **70A** causes rotation of the rotational shaft **96A** and thus causes movement of the extension shaft **96B** around the longitudinal axis of the rotational shaft **96A** as described in additional detail below.

[0080] Continuing with FIGS. **2B**, **2D**, and **2F**, the extension shaft **96B** is coupled to or integral with the rotational shaft **96A** and extends radially from the rotational shaft **96A** such that the longitudinal axis of the extension shaft **96B** is transverse to the longitudinal axis of the rotational

shaft **96A**, thereby resulting in the T-shaped configuration of the T shaft **96**. When the pitch shaft **96A** is rotatably disposed within the yoke opening **93** as described above, the extension shaft **96B** extends from the rotational shaft **96A** as best shown in FIG. 2F such that the extension shaft **96B** is disposed in either of the shaft slots **94A**, **94B** between the prong sets **92A**, **92B** depending on the rotational position of the rotational shaft **96A**. That is, as the rotational shaft **96A** rotates, the extension shaft **96B** rotates around the longitudinal axis of the rotational shaft **96A** along a path that includes the two shaft slots **94A**, **94B** such that the extension shaft **96B** can rotate at least 180° around the rotational shaft **96A**.

[0081] Continuing with FIG. 2F, the shoulder joint **46** also has a second shoulder bevel gear **104** rotatably disposed on the end of the rotational shaft **96A** opposite the first shoulder bevel gear **100** such that the second shoulder gear **104** can rotate in relation to the rotational shaft **96A**. The second shoulder bevel gear **104** is rotatably coupled to the inner drive bevel gear **56B** such that rotation of the inner drive bevel gear **56B** causes rotation of the second shoulder bevel gear **104**. Further, the second shoulder bevel gear **104** is also rotatably coupled to the output body (also referred to as the “roll body”) bevel gear **108** such that rotation of the second shoulder gear **104** causes rotation of the output body bevel gear **108**. As such, rotation of the inner drive bevel gear **56B** causes rotation of the output body **106**, as described in additional detail below.

[0082] The output body **106** with the output body bevel gear **108** is rotatably disposed on the extension shaft **96B** such that the output gear **108** and body **106** can rotate in relation to the extension shaft **96B**. Thus, rotation of the output body bevel gear **108** around the extension shaft **96B** caused by rotation of the second shoulder bevel gear **104** also causes rotation of the output body **106**. Further, rotation of the extension shaft **96B** around the rotational axis of the rotational shaft **96A** causes both the extension shaft **96B** and the output body **106** disposed thereon to also rotate around the rotational axis of the shaft **96A**, thereby resulting in the pitch motion of the arm attached thereto (such as arm **130** or upper arm **150**, for example).

[0083] In use, the outer drive bevel gear **56A** is rotatably coupled to the first shoulder bevel gear **100** as discussed above such that rotation of the outer drive bevel gear **56A** causes rotation of the first shoulder bevel gear **100**. This rotation of the first shoulder bevel gear **100** causes rotation of the rotational shaft **96A**, which causes the movement of the extension shaft **96B**—and the output body **106** coupled thereto—radially around the longitudinal axis of the rotational shaft **96A**, which results in pitch.

[0084] Further, the inner drive bevel gear **56B** is rotatably coupled to the second shoulder bevel gear **104** as discussed above such that the rotation of the inner drive bevel gear **56B** causes rotation of the second shoulder bevel gear **104**. This rotation of the second shoulder bevel gear **104** causes rotation of the output body bevel gear **108**, which causes rotation of the output body **106** around the extension shaft **96B**, which results in roll.

[0085] In addition, the inner driveshaft **52C** is coupled to—or integral with—the differential yoke **56C** as discussed above such that rotation of the inner driveshaft **52C** causes rotation of the differential yoke **56C**, which results in yaw.

[0086] Thus, the shoulder joint **46** provides three degrees of freedom. The shoulder joint **46** configuration provides those three degrees of freedom while minimizing the cross-sectional size thereof. FIG. 2E depicts the greatest cross-sectional diameter of the shoulder joint **46**, which is also the largest cross-sectional diameter of the internal components of the device **40** and any other device embodiment disclosed or contemplated herein. As such, according to certain implementations, the cross-sectional diameter of the second or internal body (such as internal body **44**) is no greater (or not substantially greater) than the cross-sectional diameter of the shoulder joint (such as shoulder joint **46**), as best shown by example in FIG. 2B. In one implementation, the shoulder joint **46** has a maximum diameter of about 15 mm, which means that the internal body **44**, the shoulder joint **46**, and the robotic arm (not shown) coupled thereto can be inserted through any trocar or other insertion device having a minimum internal diameter of about 15 mm.

[0087] FIG. 3 depicts yet another embodiment of a shoulder joint **120** with a similar configuration to the joint **46** discussed above and a maximum diameter of about 13.5 mm, meaning that any device embodiment having such a joint **120** can have an internal body, shoulder joint **120**, and coupled robotic arm that can be inserted through any trocar or other insertion device having a minimum internal diameter of 13.5 mm.

[0088] Alternatively, any shoulder joint (such as joint **46** or **120**) according to any embodiment disclosed or contemplated herein can have a maximum diameter ranging from about 13.5 mm to any desirable greater diameter. In a further embodiment, the maximum diameter of any such shoulder joint can be about 18 mm.

[0089] In the specific embodiments of FIGS. **1A-3** and the various other implementations disclosed or contemplated herein, one advantage of the in-line, three degree-of-freedom (“3 DOF”) shoulder joint is the resulting reduced cross-sectional profile. That is, the entire device (such as device **40** or other such embodiments herein) has a smaller cross-sectional diameter (transverse to the longitudinal axes of the body and the single arm) in comparison to known devices having motors (and other such actuators) in the body thereof, including such known devices with two arms. The known two-armed devices have two arms attached to a single body such that the arms have mirrored, symmetric functionality. This known two-armed mirrored configuration results in a device with a larger cross-sectional profile. In contrast, the single-armed device embodiments disclosed or contemplated herein have smaller cross-sectional profiles and thus can be incorporated into a system of two or more such devices that allow for positioning the two or more devices in various locations into a patient's abdomen through various incisions at different locations therein. Given the variation in human anatomy across patients, the positioning of procedural tools and accessories can vary significantly by patient such that the use of separate single-armed devices can provide an advantage. Further, the use of two or more separate single-armed devices provides spatial and operational independence between the two or more robotic arms, thereby enabling independent gross positioning capabilities. As such, the device (such as device **40** or any other embodiment disclosed or contemplated herein) can be inserted through a smaller opening (such as a trocar port, other known port, or incision, for example) for use in a medical procedure, or two or more such devices can be inserted through two or more openings, thereby resulting in a less invasive procedure than the known devices.

[0090] Further, as discussed above, the in-line, 3 DOF shoulder joint embodiments herein (such as, for example, joints **46** or **120**) have a maximum diameter that is smaller than most separate shoulder joints in the known two-armed devices. This reduced cross-sectional diameter results from the in-line shoulder joint configuration that consists of the combination of the differential yoke (such as yoke **16C**) and the dual shaft (such as dual shaft **96**) and how those two components are positioned with the various gears within the shoulder to result in a minimal diameter. Using the shoulder joint **46** as an example, the disposition of the rotatable dual shaft **96** within the differential yoke **16C** in combination with (1) the rotatable coupling of the inner drive shaft **52C** and the differential yoke **56C**, and (2) the rotatable coupling of the outer drive bevel gear **56A** with the first shoulder bevel gear **100** and thus with the rotational shaft **96A** of the dual shaft **96** results in minimized profile of the shoulder joint **46**. That is, unlike most known shoulder joints, the radially outermost component of the shoulder joint **46** is the first shoulder bevel gear **100**. Put another way, the component driven by the first shoulder bevel gear **100** is disposed in an interior radial position in relation to the gear **100**, rather than a position that is radially outside the gear **100** in relation to the cross-sectional midpoint of the shoulder **46**. In contrast, in most known/prior shoulder joints, such an outer bevel gear would be coupled to a component disposed on the radially outer side of the bevel gear, thus resulting in a great cross-section diameter in comparison to the shoulder joints disclosed and contemplated herein.

[0091] The “in-line” or collinear feature of the 3 DOF shoulder joint **46** results from the differential yoke **56C**—with the dual shaft **96** rotatably disposed therein—being rotationally coupled to the inner

drive shaft **52C**. Thus, when the dual shaft **96** is disposed such that the rotational shaft **96A** extends away from the internal body **44** and is parallel and concentric with the longitudinal axis of the internal body **44** (such that an upper arm of a robotic arm (not shown) coupled thereto is also parallel and concentric with the longitudinal axis of the internal body **44**), the rotation of the inner drive shaft **52C** causes the directly corresponding rotation of the differential yoke **56C** (and thus directly corresponding rotation of the upper arm attached thereto).

[0092] In certain embodiments, as best shown in FIG. **4**, a single-arm device **130** similar to the embodiments above has an external body (not shown), an internal body **132**, a shoulder joint **134**, and a single robotic arm **136** coupled to the single shoulder joint **134**. Further, in accordance with some implementations, the arm **136** has two components: a first arm component (also referred to herein as an “upper arm”) **136A** and a second arm component (also referred to herein as a “forearm”) **136B**. Further, some embodiments have an operational component (also referred to herein as an “end effector”) **138** coupled to the arm **136**.

[0093] FIGS. **5A** and **5B** depict one embodiment of an upper arm **150**, according to one embodiment. This upper arm **150** embodiment can be fixedly coupled to the output body (such as output body **106** discussed above) of the joint (such as joint **46**) such that the roll of the output body causes the upper arm **150** to rotate around its longitudinal axis. More specifically, the arm **150** has an opening **152** defined in its proximal end into which the output body (such as output body **106**) can be disposed and thereby attached to the arm **150**. Alternatively, any coupling mechanism, feature, or method can be used to couple the upper arm **150** to the output body.

[0094] The upper arm **150** also has an arm joint (also referred to herein as a “elbow joint”) **154** at its distal end that allows for a jointed coupling to a forearm (such as forearm **180** as discussed below). The elbow joint **154** has two opposing supports (or “projections”) **156A**, **156B** extending from or at the distal end of the upper arm **150** that define a space therebetween in which a coupling body **158** is rotatably disposed. Each of the supports **156A**, **156B** has a bearing **160A**, **160B** disposed on an inner wall thereof. The coupling body **158** has two rotational projections **162A**, **162B**, a coupling component **164**, and a body bevel gear **166**. The two rotational projections **162A**, **162B**, according to one implementation, are disposed within and rotatably coupled to the two bearings **160A**, **160B**, respectively, as best shown in FIG. **5B**, such that the coupling body **158** can rotate in relation to the supports **156A**, **156B**. In this embodiment, the coupling component **164** is an opening **164** defined in the coupling body **158** that is configured to receive a portion of a forearm (such as forearm **180**) such that the forearm portion is disposed within the opening **164** and the forearm is thereby coupled to the coupling body **158**. Alternatively, the coupling component **164** can be any coupling mechanism, feature, or method can be used to couple the upper arm **150** to the forearm.

[0095] Further, the joint **154** includes an elbow joint bevel gear **168** that is coupled or rotationally constrained to an upper arm motor **170** such that actuation of the motor **170** causes rotation of the elbow joint bevel gear **168**. The elbow joint bevel gear **168** is rotatably coupled to the body bevel gear **166** such that rotation of the elbow joint bevel gear **168** causes rotation of the body bevel gear **166**, which thereby causes the coupling body **158** to rotate in relation to the supports **156A**, **156B**, thereby causing the forearm (such as forearm **180**) coupled thereto to rotate in relation to the upper arm **150**. Thus, actuation of the upper arm motor **170** can cause the forearm (such as forearm **180**) to rotate in relation to the upper arm **150** at the elbow joint **154**.

[0096] In one implementation, the upper arm **150** has additional space within the arm **150** (beyond the space occupied by the upper arm motor **170**) to house additional components as necessary. For example, in one embodiment, one or more controllers (not shown) for controlling various components—including motors—of the arm or the robotic device can be disposed within the upper arm **150**.

[0097] FIGS. **6A-6C** depict one embodiment of an forearm **180**, according to one embodiment. This forearm **180** embodiment can be fixedly coupled to the coupling component **164** of the

coupling body **158** at the coupling projection **188** at the proximal end of the forearm **180**. It is understood that any coupling mechanism or method can be used for this attachment.

[0098] The forearm **180** in this implementation has a wrist joint **182** at the distal end thereof, with an operational component (“end effector”) **184** coupled to the wrist joint **182**. In this specific embodiment, the end effector **184** is a grasper **184**, but it is understood that the end effector **184** is removable and thus various known, interchangeable end effectors **184** can be used herewith. Further, as best shown in FIGS. **6B** and **6C**, the forearm **180** in this embodiment has three motors disposed within the body **190** of the forearm **180**: the first motor **186A**, the second motor **186B**, and the third motor **186C**.

[0099] One embodiment of the wrist joint **182** is depicted in additional detail in FIGS. **7A** and **7B**. More specifically, FIG. **7B** depicts a cross-sectional front view of the distal portion of the forearm body **190** and the wrist joint **182** in which certain internal components are visible, according to one exemplary embodiment. The forearm body **190** has a set of nested driveshafts **192** rotatably disposed within the body **190** that extend into the wrist joint **182**. The set of nested driveshafts **192** is made up of a first or outer driveshaft **192A**, a second or middle driveshaft **192B**, and a third or inner driveshaft **192C**. The set of nested driveshafts **192** extend from the forearm body **190** into the wrist joint **182** as shown. The inner driveshaft **192C** is rotatably disposed within the middle driveshaft **192B** as shown, and has a driven gear **194C** fixedly or integrally attached at its proximal end. At its distal end, the inner driveshaft **192C** is rotatably coupled to a first or inner drive bevel gear **196C**. The middle driveshaft **192B** is rotatably disposed within the outer driveshaft **192A** as shown, and has a driven gear **194B** fixedly or integrally attached at its proximal end. At its distal end, the middle driveshaft **192B** is coupled to a second or middle drive bevel gear **196B**. The outer driveshaft **192A** is rotatably disposed within the forearm body **190** and wrist **182** and has a driven gear **194A** fixedly or integrally attached at its proximal end. At its distal end, the outer driveshaft **192A** is coupled to a first or outer drive bevel gear **196A**.

[0100] Continuing with FIG. **7B**, the proximal end of the inner driveshaft **192C** is rotatably supported in the forearm body **190** via a first shaft bearing **198** and a second shaft bearing **200**. Further, the proximal end of the middle driveshaft **192B**, including the driven gear **194B**, is rotatably supported via the second shaft bearing **200** and a third shaft bearing **202**. In addition, the proximal end of the outer driveshaft **192A**, including driven gear **194A**, is rotatably supported via the third shaft bearing **202**.

[0101] The set of nested driveshafts **192** has three motors operably coupled thereto. More specifically, motor **186B** (of FIG. **6C**) has an motor drive gear **204A** (shown in FIG. **7A**, but not visible in FIG. **7B**) that is coupled to the driven gear **194A** (which is coupled to the outer driveshaft **192A**). Further, motor **186A** (of FIG. **6C**) has a motor drive gear **204C** that is coupled to the driven gear **194C** (which is coupled to the inner driveshaft **192C**). In addition, motor **186C** (of FIG. **6C**) has a motor drive gear **204B** (also not visible in the figures) that is coupled to the driven gear **194B** (which is coupled to the middle driveshaft **192B**).

[0102] Thus, in operation, the motor **186B** can be actuated to drive rotation of the outer driveshaft **192A** by driving rotation of motor drive gear **204A**, which drives rotation of the driven gear **194A**. Similarly, the motor **186C** can be actuated to drive rotation of the middle driveshaft **192B** by driving rotation of motor drive gear **204B** (also not visible), which drives rotation of the driven gear **194B**. In a similar fashion, the motor **186A** can be actuated to drive rotation of the inner driveshaft **192C** by driving rotation of motor drive gear **204C**, which drives rotation of the driven gear **194C**.

[0103] At the distal end, the outer driveshaft **192A** and outer drive bevel gear **196A** are supported by the first wrist bearing **210**. In addition, the middle driveshaft **192B** and the middle drive bevel gear **196B** are supported by the first wrist bearing **210** and the second wrist bearing **212**.

[0104] The inner drive bevel gear **196C** is rotatably coupled to a differential yoke **220** having two rotational projections **222A**, **222B**, an extension shaft **224** that extends into the end effector body

244, and a yoke bevel gear **226**. The differential yoke **220** is similar to the differential yoke **56C** of the shoulder joint **46** discussed above and thus conveys the same or similar in-line benefits to those of the yoke **56C**, including the minimal cross-sectional profile and the in-line configuration as discussed in detail above. The inner drive bevel gear **196C** rotatably couples to the yoke bevel gear **226**. As such, rotation of the inner drive bevel gear **196C** causes the rotation of the differential yoke **220** around the rotational projections **222A**, **222B** such that the extension shaft **224** is caused to rotate around the longitudinal axis of the yoke **220** that extends from the first rotational projection **222A** to the second **222B**.

[0105] The middle drive bevel gear **196B** is rotatably coupled to a first wrist bevel gear **230**, which is rotatably disposed over the first inner rotational projection **222A**. In addition, the first wrist bevel gear **230** is also rotatably disposed over the first outer rotational projection **246A** extending from the first wrist support bracket **246**. The first wrist support bracket **246** extends from the distal end of the forearm body **190**. As such, the first wrist bevel gear **230** is rotatably supported by the first inner rotational projection **222A** and the first outer rotational projection **246A**. Further, the first wrist bevel gear **230** is rotatably coupled to the end effector output bevel gear **232**, which is coupled to or integral with the end effector output body **234**. Thus, rotation of the middle drive bevel gear **196B** causes rotation of the first wrist bevel gear **230**, which causes rotation of the end effector output bevel gear **232**, which causes rotation of the end effector output body **234**. In certain embodiments, the output body **234** is operably coupled to the end effector to actuate the end effector in some fashion, depending on the type of end effector.

[0106] The outer drive bevel gear **196A** is rotatably coupled to the second wrist bevel gear **240**, which is rotatably disposed over the second inner rotational projection **222B**. In addition, the second wrist bevel gear **240** is also rotatably disposed over the second outer rotational projection **248A** extending from the second wrist support bracket **248**. The second wrist support bracket **248** extends from the distal end of the forearm body **190** in a fashion similar to the first wrist support bracket **246**. As such, the second wrist bevel gear **240** is rotatably supported by the second inner rotational projection **222B** and the second outer rotational projection **248A**. Further, the second wrist bevel gear **240** is rotatably coupled to the end effector rotation bevel gear **242**, which is coupled to or integral with the end effector body **244**. Thus, rotation of the outer drive bevel gear **196A** causes rotation of the second wrist bevel gear **240**, which causes rotation of the end effector rotation bevel gear **242**, which causes rotation of the end effector body **244**, thereby causing rotation of the end effector around its longitudinal axis.

[0107] The wrist joint embodiment 182 discussed above is a compact dexterous wrist joint. Given the unpredictable nature of a patient's anatomy, the criticality of a robotic device providing as much control as possible to a surgeon for the purpose of completing a surgical task, and the importance of tool dexterity in surgical procedures, the dexterity of the wrist joint in a robotic arm is important. There are known devices that provide a wrist joint with seven degrees of freedom, but those devices have arm and end effector actuation via actuators (which are typically motors) disposed outside the patient's body, such as, for example, the EndoWrist® of the Da Vinci® system. The use of mechanical (cables, etc.), pneumatic, hydraulic, or other such force transmission systems make it possible for the EndoWrist and other such devices to maintain a smaller profile while having great dexterity. In contrast, to date, known devices having actuators (such as motors) disposed within the patient's body or the opening through which the device is positioned have not been able to approach that number of degrees of freedom or the corresponding amount of dexterity. However, the wrist embodiments (such as wrist joint **182**) provide additional dexterity in comparison to other known devices with internal actuators. That is, the wrist joint **182** provides three degrees of freedom while minimizing its cross-sectional profile due to the joint **182** configuration. More specifically, as mentioned above, the configuration of the wrist joint **182** with the differential yoke **220** that is similar to the differential yoke **56C** of the shoulder joint **46** results in similar features and functionality, including the minimal cross-sectional diameter and the in-line configuration.

[0108] FIGS. 8A and 8B depict one example of a trocar **250** according to one embodiment that is sized and configured to allow for receiving therethrough the various robotic device embodiments disclosed or contemplated herein. The trocar **250** has a port (or “seal”) **252** and a cannula **254** extending from the port **252**. The port **252** and cannula **254** define an inner lumen **256** having a diameter that is large enough to receive any arm and internal body of any device implementation disclosed or contemplated herein. Further, the cannula **254** has a length such that the shoulder joint of the robotic device positioned therethrough extends through and is positioned distally out of the distal end **258** of the cannula **254**, thereby allowing the arm coupled to the shoulder joint to move in any direction as desired without being hindered or obstructed by the cannula **254**. As also discussed elsewhere herein, in addition to this trocar **250** and any other known laparoscopic trocar, it is understood that any insertion device of the appropriate size (including any type of port) or any incision of the appropriate size can be used with the various device embodiments disclosed or contemplated herein.

[0109] Trocar ports such as the trocar **250** described above and other known trocars can have certain advantages over known single-site entry ports. For example, insertion of certain known robotic devices through the single-site entry port can cause structural damage to the robotic device as a result of the physical contact between the port and the device. In addition, such known single-site entry ports require a larger incision in the patient than standard trocars require.

[0110] In use, one embodiment of a single-arm robotic device **270** is depicted in FIGS. **9A** and **9B** in a surgical environment. More specifically, FIG. **9A** provides an external view of the device **270** positioned in a patient **272** through a trocar **274** such that the device **270** is in its operational configuration. That is, the device **270** is positioned into the target cavity (not shown) of the patient **272** such that the robotic arm (not shown) can be used to perform the desired procedure therein. In this embodiment, the device **270** is stabilized and/or maintained in the desired operational position via a known external support **276** that is coupled to the external body **278** of the device **270**.

[0111] Further, FIG. **9B** is a schematic view of the device **270** and the visual representation of the workspace **280** in which the device **270** can operate. More specifically, the workspace **280** is the visual representation of the full area that the end effector **284** coupled to the single arm **282** can reach. As such, the device **270** can be positioned within a target cavity in a patient such as the patient **272** in FIG. **9B** such that the interior body **286** extends into the target cavity, resulting in the robotic arm **282** being disposed therein. In this specific embodiment, the device **270**, which has components and functionality similar to the various other device embodiments disclosed or contemplated herein, can manipulate or otherwise position the arm **282** such that the end effector **284** can extend to all of the areas defined by the workspace **280**. Thus, the end effector **284** can be used to perform a procedure at any location within that workspace **280**.

[0112] Although the present invention has been described with reference to preferred embodiments, persons skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the invention.

Claims

1. A robotic device comprising: (a) a device body comprising: (i) a proximal section; and (ii) a distal section; (b) a first driveshaft operably coupled to a first motor; (c) a second driveshaft operably coupled to a second motor; (d) a first bevel gear operably coupled to the second driveshaft at a distal end of the second driveshaft; (e) a shoulder comprising: (i) a differential yoke rotationally coupled to the first driveshaft, the differential yoke comprising: (A) a yoke body having a longitudinal axis and configured to rotate about a first rotational axis; and (B) a pair of curved shells extending therefrom and defining a yoke lumen having a longitudinal axis that is transverse to the longitudinal axis of the yoke body; and (ii) a dual shaft rotatably disposed within the yoke lumen, the dual shaft comprising: (A) a rotational shaft rotatably disposed within the yoke

lumen, the rotational shaft rotationally coupled to the first bevel gear, wherein the rotational shaft rotates about a second rotational axis; and (B) an extension shaft extending from the rotational shaft such that a longitudinal axis of the extension shaft is transverse to the second rotational axis; and (iii) an output body rotatably disposed around the extension shaft such that the extension shaft is rotatably disposed within a lumen of the output body, and wherein the output body rotates about a third rotational axis, wherein the first, second, and third rotational axes intersect at a single point.

2. The robotic device of claim 1, wherein the first and second motors are disposed within the device body.

3. The robotic device of claim 1, wherein the extension shaft is fixedly attached to the rotational shaft.

4. The robotic device of claim 1, further comprising a third driveshaft operably coupled to a third motor, the third driveshaft operably coupled at a distal end to a second bevel gear, wherein the second bevel gear is operably coupled to the output body.

5. The robotic device of claim 1, further comprising an arm operably coupled to the output body.

6. The robotic device of claim 1, wherein the shoulder is a unitary shoulder joint.

7. The robotic device of claim 1, wherein the proximal section has a diameter, and the distal section has a diameter that is less than the diameter of the proximal section.

8. A robotic device comprising: (a) a device body comprising: (i) a proximal section; and (ii) a distal section; (b) a first driveshaft operably coupled to a first motor; (c) a second driveshaft operably coupled to a second motor, the second driveshaft coaxial with the first driveshaft and operably coupled at a distal end to a first bevel gear; (d) a shoulder comprising: (i) a differential yoke rotationally coupled to the first driveshaft, the differential yoke comprising a yoke body and a yoke lumen defined in the differential yoke, wherein the lumen has a longitudinal axis that is transverse to the longitudinal axis of the yoke body; and (ii) a dual shaft rotatably disposed within the yoke lumen, the dual shaft comprising: (A) a rotational shaft rotatably disposed within the yoke lumen, the rotational shaft rotationally coupled to the first bevel gear; and (B) an extension shaft extending from the rotational shaft such that a longitudinal axis of the extension shaft is transverse to a longitudinal axis of the rotational shaft; and (iii) an output body rotatably disposed around the extension shaft such that the extension shaft is rotatably disposed within a lumen of the output body, wherein the shoulder is an in-line joint that is collinear with the distal section.

9. The robotic device of claim 8, wherein the first and second motors are disposed within the device body.

10. The robotic device of claim 8, further comprising an arm operably coupled to the output body.

11. The robotic device of claim 8, wherein the yoke lumen is defined by opposing first and second yoke lumen walls, the first yoke lumen wall comprising a first slot and the second yoke lumen wall comprising a second slot.

12. The robotic device of claim 11, wherein the proximal section has a diameter, and the distal section has a diameter that is less than the diameter of the proximal section.

13. A robotic device comprising: (a) a first driveshaft disposed through a device body and operably coupled to a first motor; (b) a second driveshaft disposed through the device body and operably coupled to a second motor, the second driveshaft operably coupled at a distal end to a first shoulder gear; (c) a third driveshaft disposed through the device body and operably coupled to a third motor, the third driveshaft operably coupled at a distal end to a second shoulder gear; and (d) a shoulder comprising: (i) a differential yoke rotationally coupled to the first driveshaft, the differential yoke defining a yoke lumen therewithin; and (ii) a T shaft rotatably disposed within the yoke lumen, the T shaft comprising: (A) a rotational shaft rotatably disposed within the yoke lumen, the rotational shaft rotationally coupled to the first shoulder gear; and (B) an extension shaft extending from the rotational shaft; (iii) an output body rotatably disposed over the extension shaft and having an output body gear, the output body gear rotationally coupled to the second shoulder gear, wherein the first shoulder gear defines a radial outer limit of the shoulder.

- 14.** The robotic device of claim 13, wherein the first, second, and third motors are disposed within the device body.
- 15.** The robotic device of claim 13, wherein the yoke lumen is defined by opposing first and second yoke lumen walls, the first yoke lumen wall comprising a first slot and the second yoke lumen wall comprising a second slot.
- 16.** The robotic device of claim 15, wherein the extension shaft is positionable within the first and second slots.
- 17.** The robotic device of claim 13, wherein a longitudinal axis of the extension shaft is transverse to a longitudinal axis of the rotational shaft.
- 18.** The robotic device of claim 13, further comprising an arm operably coupled to the output body.
- 19.** The robotic device of claim 18, wherein the arm comprises an upper arm body; a forearm body; an elbow joint coupling the forearm body to the upper arm body; an end effector; and a wrist joint coupling the end effector to the forearm body, wherein the upper arm body is operably coupled to the output body.
- 20.** The robotic device of claim 13, wherein the first driveshaft is disposed within the third driveshaft, and wherein the third driveshaft is disposed within the second driveshaft.
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